

Question Number	Answer	Additional Guidance	Mark
2(a)	there will be no difference between the biomass of Almo and Kanlow (switchgrass)	Accept no difference in biomass between the two varieties of switchgrass	(1) exp

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	the organic matter of living organisms in a given area	Accept biological material for organic matter Accept organic matter per area per unit time eg per hectare per year	(1) exp

Question Number	Answer	Additional Guidance	Mark
2(b)(ii)	- correct calculation of energy content (1) - answer expressed to two significant figures in standard form (1)	162 450 MJ 1.6×10^5 MJ ALLOW consequential error for MP2 Correct answer with no working gets 2 marks	(2) exp

Question Number	Answer	Additional Guidance	Mark																											
2(b)(iii)	- suitable table format, with data, correct column headings and units (1) - medians correctly identified as 14.0 and 15.2 (1)	Do not accept units in the body of the table. eg. <table><tr><th colspan="3">Biomass (of switchgrass) x10³ kg / hectare</th></tr><tr><th>year</th><th>Almo</th><th>Kanlow</th></tr><tr><td>1</td><td>20.1</td><td>15.5</td></tr><tr><td>2</td><td>19.2</td><td>22.2</td></tr><tr><td>3</td><td>14.2</td><td>14.4</td></tr><tr><td>4</td><td>11.4</td><td>11.4</td></tr><tr><td>5</td><td>11.8</td><td>12.6</td></tr><tr><td>6</td><td>13.6</td><td>15.2</td></tr><tr><td>7</td><td>14.0</td><td>16.5</td></tr></table>	Biomass (of switchgrass) x10 ³ kg / hectare			year	Almo	Kanlow	1	20.1	15.5	2	19.2	22.2	3	14.2	14.4	4	11.4	11.4	5	11.8	12.6	6	13.6	15.2	7	14.0	16.5	(2) exp
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year	Almo	Kanlow																												
1	20.1	15.5																												
2	19.2	22.2																												
3	14.2	14.4																												
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5	11.8	12.6																												
6	13.6	15.2																												
7	14.0	16.5																												

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2(c)	- bar graph with linear scale, correct axis labels with units (1) - medians plotted correctly (1) - range bars plotted correctly (1)	Scale must start at zero Allow ECF from 2biii	(3) exp

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	- correct substitution of given R values in the equation (1) - correct value of U_2 (1)	19.5 Correct answer with no working gets 2 marks	(2) exp

Question Number	Answer	Additional Guidance	Mark
2(d)(ii)	An answer that includes three of the following points: - the calculated value of U (19.5) is more than the critical value which is 8 (at $p = 0.05$) (1) - therefore accept the null hypothesis - there is no difference between the yield of Almo and Kanlow varieties of switchgrass (1) - comment on variability of data (1)	Allow ECF from (d)(i) Accept critical value circled in table eg range bars overlap	(3) exp

Question Number	Answer	Additional Guidance	Mark
2(e)	An answer that includes three of the following points: - only one field was used so it may not be a representative sample (for biomass from these varieties) (1) - the plots for each variety may not have been randomly selected (1) - one named environmental condition may not be the same for each variety (1) - comment on the range of data over the seven-year period (1)	Do not accept small sample size Accept eg. all of the Almo plots may have been on one side of the field and all of the Kanlow plots on the other eg. shading, soil type, soil moisture, mineral content, slope of field eg. number of growing days / weather conditions might not be the same each year	(3) exp

(Total for question 2 = 17 marks)

